

0 4

Scientists investigated the effect of different types of animal farming on the diversity and number of dung beetles. They determined the number of dung beetle species and their total number on intensive (I), rough grazing (R) and organic (O) farms.

Figure 4 and **Figure 5** show some of their results.

Figure 4

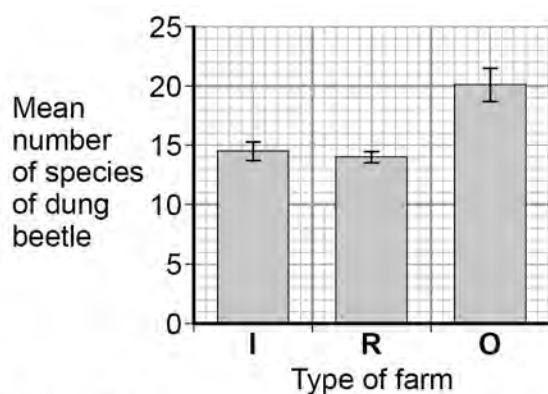
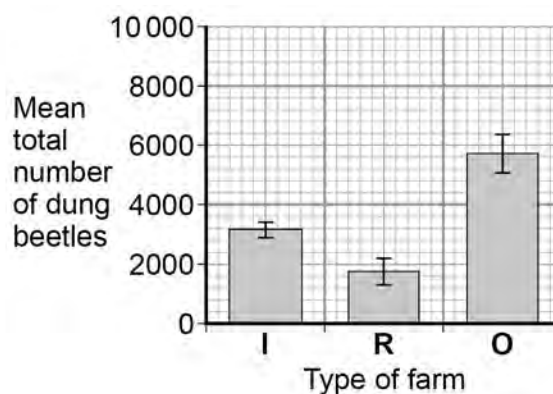


Figure 5



Key: I Standard deviation

0 4 . 1

What is the mean species richness for dung beetles on the rough grazing farms? **[1 mark]**

0 4 . 2

In addition to the information provided in **Figures 4** and **5**, what other measurement is required to calculate an index of diversity for dung beetles? **[1 mark]**

0 4 . 3

Explain what the standard deviations suggest about the difference in mean total number of dung beetles between the different types of farm. **[2 marks]**



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The scientists placed traps to collect the dung beetles at sites chosen at random.

Explain the importance of the sites being chosen at random.

[1 mark]

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On the intensive farms, the farmers had removed hedges to increase land for grazing. This resulted in a decrease in the diversity of birds on these farms.

Explain why the removal of hedges caused a decrease in the diversity of birds.

[3 marks]

Turn over for the next question

